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EXAMINER	
KWIATKOWSKA, LIDIA	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/244,379	Applicant(s) Gao et al.	
	Examiner LIDIA KWIATKOWSKA	Art Unit 3666	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 29 April 2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 29 April 2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 1-20 are rejected under 35 U.S.C. 103 as being unpatentable over **Dyer (US Patent No. US20210053567A1)** in view of **Kaufman (US Patent No. US20200310461A1).**

Regarding claim 1, Dyer teaches, a method of determining a stopping location for a vehicle before picking up a passenger, the method comprising: by one or more processors, at least one of which is a component of an autonomous vehicle (AV); (**See Dyer paragraph 0002;** "One aspect of the disclosure provides a method of maneuvering a vehicle in an autonomous driving mode. The method includes identifying, by the one or more processors, a time constraint for a pullover maneuver; identifying, by the one or more processors, a geographic constraint for the pullover maneuver; inputting, by the one or more processors, the time constraint and the geographic constraint into a model in order to receive a list of qualities for a region, the region including a plurality of pullover locations; determine, by the one or more processors, whether to attempt to find a pullover location within the region to perform the pullover maneuver based on the list of qualities for the region; and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*);

receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device; determining a pickup area for the ride service request, along with

a loading point that is within a pickup area; (**See Dyer paragraph 0042;** "Although the client computing devices **420**, **430**, and **440** may each comprise a full-sized personal computing device, they may alternatively comprise mobile computing devices capable of wirelessly exchanging data with a server over a network such as the Internet. By way of example only, client computing device **420** may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device **430** may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph 0076;** " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices **110** may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger.");

determining a route from a starting location of the AV to the pickup area; (**See Dyer paragraph 0027;** "Planning system **168** may be used by computing devices **110** in order to determine and follow a route generated by a routing system **166** to a location. For instance, the routing system **166** may use map information to determine a route from a current location of the vehicle to a drop off location. ").

navigating the AV along the route toward the pickup area; (**See Dyer paragraph 0036;** "The computing devices **110** may control the vehicle in an autonomous driving mode by controlling

various components. For instance, by way of example, the computing devices **110** may navigate the vehicle to a destination location completely autonomously using data from the detailed map information and planning system **168**. ");

Dyer does not explicitly teach but Kaufman teaches, before the AV reaches the pickup area, determining whether the one or more processors have received a departure confirmation from the passenger electronic device, wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point; (**See Kaufman paragraph 0032;** "The AV coordination component 138 receives the information in the ride request, and in response to receiving such information, the AV coordination component 138 determines a route toward the pickup location of the person. In another example, the server coordination component 114 can determine the route and transmit the route to the autonomous vehicle 106. Responsive to the route being determined, the autonomous vehicle 106 travels along the route toward the pickup location of the person. As the autonomous vehicle 106 travels along the route, the autonomous vehicle 106 can ascertain pickup refinement information for the person. In one example, pickup refinement information can be based on strength of a wireless signal emanating from the mobile computing device 104 as the autonomous vehicle 106 moves relative to the mobile computing device 104 (e.g., as the signal strength becomes stronger, the autonomous vehicle 106 can ascertain that it is getting closer to the person). In another example, the autonomous vehicle 106 can ascertain the pickup refinement information based upon a passenger identifier being displayed on the display 130 of the mobile computing device 104. In yet another example, the autonomous vehicle 106 can

ascertain the pickup refinement information based upon a passenger identifier in the form of physical object being held in the air by the person. Other examples are also contemplated. "; also **see Kaufman paragraph 0052;** "The server computing system processes the ride request and transmits ride information to an autonomous vehicle that is available to pick up the person at the pickup location identified in the ride request. The mobile computing device can confirmation that the ride request has been accepted by the server computing system."); and if the one or more processors have received the departure confirmation, causing the AV to navigate into the pickup area and stop at the loading point, otherwise causing the AV to either (a) navigate to an intermediate stopping location before reaching the pickup area or (b) pass through the pickup area.; (**see Kaufman paragraph 0033;** "As the autonomous vehicle 106 continues to navigate proximate the pickup location identified in the ride request, the sensor (not shown) of the autonomous vehicle 106 scans the environment. When the autonomous vehicle 106 detects the passenger identifier based upon a sensor signal output by the sensor of the autonomous vehicle 106, the autonomous vehicle 106 ascertains identifying information based upon the passenger identifier (e.g., the autonomous vehicle 106 reads a QR code and acquires the identifying information). The autonomous vehicle 106 compares the identifying information to the identifying information received from the server computing system 102. If the two pieces of identifying information do not match, the autonomous vehicle 106 continues to search for the passenger identifier. If the two pieces of identifying information match, the autonomous vehicle 106 maneuvers towards the person presenting the passenger identifier and allows the person entry into the autonomous vehicle 106").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Koufman request confirmation and passenger identifier. No new functionality would arise from the combination and the

combination would improve usability of Dyer by adding the confirmation and passenger identifier. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 2, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer does not explicitly teach, but Kaufman teaches, wherein determining the loading point comprises: determining a passenger identifier that is associated with the passenger electronic device; accessing a passenger profile server and querying the passenger profile server for a passenger profile that corresponds to the passenger identifier; accessing the passenger profile to extract one or more loading point location rules from the passenger profile; and using th e loading point location rules to select the loading point from a set of candidate loading points (**see Kaufman paragraph 0026;** "The server computing system 102 is configured to receive a ride request from the mobile computing device 104, wherein the ride request comprises an identifier for a passenger account of a person and ride information, and further wherein the ride information can include a pickup time, a pickup location, a ride destination, a number of passengers who are to be included in the ride, etc. In response to receiving the ride request, the server computing system 102 is configured to transmit a signal to the autonomous vehicle 106,

wherein the signal comprises a ride instruction. The ride instruction can indicate at least some of the content of the ride request, including the ride information and a passenger identifier that is assigned to the person. The passenger identifier may be a barcode (e.g., a QR code), a pattern or a sequence of patterns, a color or a sequence of colors, a key chain (with unique shape, color, a wireless transceiver, etc.). In this example, the autonomous vehicle 106 conducts a search for the passenger identifier while the autonomous vehicle 106 travels through a search area that is proximate to the pickup location identified in the ride request."; as well **as see Kaufman paragraph 0054;** "With reference now to FIG. 5, an exemplary methodology 500 performed by a server computing system is illustrated, wherein the server computing system is configured to instruct an autonomous vehicle to pickup a person and transport the person to a desired destination of the person. The methodology 500 starts at 502, and at 504 the computing system receives a ride request from a mobile computing device, wherein the ride request comprises an identifier of the person, a pickup location, and a destination. At 506, the server computing system transmits ride instructions to an available autonomous vehicle from among a fleet of autonomous vehicles that are in network communication with the server computing system. The ride instructions can include, for instance, identifying information for the person (which is assigned to a passenger identifier), a pickup location, a pickup time, a destination location, etc. The server computing system can identify the autonomous vehicle based upon preferences in a passenger account of the person, based upon a number of riders indicated in the ride request, etc.").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Koufman confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 3, Modified Dyer teaches, the method of claim 2 as discussed above. Dyer further teaches, wherein the loading point location rules comprise one or more rules to prioritize one or more of the following characteristics to determine the loading point: a lane segment that is in front of an entry of a building having a location that corresponds to a location of the passenger electronic device; a lane segment that is midway between two cross streets; or a lane segment that is as close as possible to the cross street that is next in the direction of traffic while remaining a threshold distance away from that cross street; (**see Dyer paragraph 0073 – 0075**; "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may

do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.

In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).

In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 4, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches, further comprising, by the one or more processors, determining the intermediate stopping location as a location that: is along the route between a current location of the AV and the loading point; and satisfies each of a plurality of permissible stopping location criteria (**See Dyer paragraph 0075;** "In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities *(Examiner notes; list of qualities can also be a plurality of permissible stopping location criteria)* or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 5, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches, further comprising, by the one or more processors after the one or more processors have received the departure confirmation: before the AV reaches the loading point, processing data received from a perception system of the AV to determine whether the AV can

stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV; and only stopping the AV at the loading point to perform the loading operation if the loading operation will not obstruct the another vehicle, otherwise navigating the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location (**see Dyer paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)."; also **see Dyer paragraph 0073;** " In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.").

Regarding claim 6, Modified Dyer teaches, the method of claim 5 as discussed above. Dyer further teaches, wherein determining whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV comprises: determining whether the loading point is in a location that will require the AV to at least partially block a lane in the other vehicle that is following the AV is currently traveling; and determining that the AV will obstruct the other vehicle if: the loading point is in a location that will require the AV to at least partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop;(**see Dyer paragraph 0033;** "The perception system 172 also includes one or more components for detecting objects external to the vehicle such as other vehicles, obstacles in the roadway, traffic signals, signs, trees, etc.…."; also **see Dyer paragraph 0045;** "Other signals stored in the storage system 450 may also be determined by the server computing devices and/or human operators for areas around such pullover locations, or the regions discussed above. These other signals may include the number of available pullover locations (as the vehicle drives through the region), how congested such areas are, how many other vehicles are pulling over in the region (as the vehicle drives through the region), how many vehicles are double parked within the region, whether there are other types of road users (such as pedestrians or cyclists) within the region, etc."; further **see Dyer paragraph 0046;** "Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle *(Examiner notes- tracking delta between the other road user's that would include vehicles behind the AV)*. This vehicle inconvenience value may be

determined from prior attempts to pullover which may have caused problems to other road users *(Examiner notes – pullover causing problems to other users can include: partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop)* .For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience).

Regarding claim 7, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer does not teach but Kaufman teaches; further comprising, by the one or more processors after the one or more processors have received the departure confirmation and before the AV reaches the loading point, processing data received from a perception system of the AV to; (**see Kaufman paragraph 0055;** "At 508, the server computing system receives confirmation from the autonomous vehicle that the ride instructions have been received. At 510, the server computing system transmits a passenger identifier (e.g., a barcode, a color, a pattern, etc.) to the mobile computing device if the mobile computing device does not already have the passenger identifier. In addition, the server computing system can transmit instructions to the mobile computing device, wherein the instructions indicate when the person is to present the passenger identifier and a direction that the passenger identifier is to be presented. At 512, the server computing system receives an indication that the autonomous vehicle has successfully located the passenger *(Examiner notes; location of the passenger is equivalent to that receiving the confirmation of departure)* and the ride has begun. The methodology 500 completes at 510.")

identify one or more objects located proximate to the loading point; measure a length of a stop zone that includes the loading point and that is free from obstruction; use a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone in compliance with the permissible stopping criteria; and only stop at the loading point to perform the loading operation if the AV can non-suddenly stop within the stop zone in compliance with the permissible stopping criteria, otherwise navigate the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location (**see Kaufman paragraph 0010;** "In another example, the person may not have ordered a ride through the application (e.g., the person's mobile telephone is dead, or the person's mobile telephone does not get service). The person leaves the basketball arena and presents his or her mobile telephone to an autonomous vehicle that is not currently providing a ride, is not in route to pick up a person who has hailed a ride, and/or which lacks capacity for the person (e.g., the autonomous vehicle already has several riders). The autonomous vehicle can detect the passenger identifier for the person from among the other visual stimuli in the surrounding environment based upon a sensor system output by a sensor of the autonomous vehicle. The computing system of the autonomous vehicle can identify a passenger account based upon the passenger identifier, and the autonomous vehicle may then stop to pick up the person and transport the person to a destination location indicated by the person."; also **see Kaufman paragraph 0033;** "As the autonomous vehicle 106 continues to navigate proximate the pickup location identified in the ride request, the sensor (not shown) of the autonomous vehicle 106 scans the environment. When the autonomous vehicle 106 detects the passenger identifier

based upon a sensor signal output by the sensor of the autonomous vehicle 106, the autonomous vehicle 106 ascertains identifying information based upon the passenger identifier (e.g., the autonomous vehicle 106 reads a QR code and acquires the identifying information). The autonomous vehicle 106 compares the identifying information to the identifying information received from the server computing system 102. If the two pieces of identifying information do not match, the autonomous vehicle 106 continues to search for the passenger identifier. If the two pieces of identifying information match, the autonomous vehicle 106 maneuvers towards the person presenting the passenger identifier and allows the person entry into the autonomous vehicle 106").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Koufman request confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the request confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 8, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches, further comprising, by the one or more processors after the AV reaches the pickup area: receiving a nudge request; processing data received by a perception system of the

AV to determine whether any objects would obstruct the AV if the AV would move to a position that is ahead of the loading by a specified distance; and if no objects would obstruct the AV, update the loading point to move forward by the specified distance and move the vehicle to the updated loading point; (**see Dyer paragraph 0046;** "Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. Such signals may include how long passengers take to arrive, board and depart at a pullover location, the passenger inconvenience value of a pullover location, the vehicle inconvenience at pullover location, how long the vehicle is able to stay in a pullover location, etc. The values for passenger inconvenience and vehicle inconvenience may be determined, for instance, on a scale of 0 to 1 using a model which generates such values given map data and sensor data collected during such attempted pullovers. Passenger inconvenience values may represent how convenient a particular pickup up or drop off was for a passenger by measuring how much extra distance is imposed on the passenger by the selection of a particular pullover location. Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle. This vehicle inconvenience value may be determined from prior attempts to pullover which may have caused problems to other road users. For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience value.").

Regarding claim 9, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches, further comprising, by the one or more processors when the one or more processors have not received the departure confirmation, determining the intermediate stopping location by: selecting a road segment that is a nearest useable road segment to the pickup location; identifying, from one or more permissible stopping points along the road segment, the permissible stopping point that is closest to the pickup location; and using the permissible stopping point that is closest to the pickup location as the intermediate stopping location; (**See Dyer Paragraph 74;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)." ; **also see Dyer paragraph 0046;**"Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. "; as well **see Dyer paragraph 0048;** "In order to build a model, the server computing devices 410 may access the signals of the storage system 450 described above. Using these signals or values determined from these signals, a model that identifies expected characteristics of regions may be built by the server computing devices 410. The model may be trained such that for a given geographic constraint (such as a specific location or a region) and time constraint (such as day of the year, calendar month, day

of week, and/or time of day), the model may provide a list of expected qualities for a region. In this regard, if a specific location is provided, the region may correspond to a region that includes the specific location. In this regard, the signals or values determined from these signals for a given region where the sensor data corresponding to the signals was collected may be used as training outputs for the model, and the specific location or given region and a time when the sensor data was collected may be used as training inputs to the model. In the case of a machine-learned model, the training may essentially tune parameter values for the model.").

Regarding claim 10, Dyer teaches, a ride service system, comprising: an autonomous vehicle (AV) comprising a perception system, a motion planning system and a motion control system; a processor; and a memory containing programming instructions that are configured to cause the processor to determine a stopping location for the AV before picking up a passenger by; (**See Dyer paragraph 0025**; " In one aspect the computing devices 110 may be part of an autonomous control system capable of communicating with various components of the vehicle in order to control the vehicle in an autonomous driving mode. For example, returning to FIG. 1, the computing devices 110 may be in communication with various systems of vehicle 100, such as deceleration system 160, acceleration system 162, steering system 164, routing system 166, planning system 168, positioning system 170, and perception system 172 in order to control the movement, speed, etc. of vehicle 100 in accordance with the instructions 134 of memory 130 in the autonomous driving mode."; also **See Dyer paragraph 0002;** "determine, by the one or more processors, whether to attempt to find a pullover location within the region to perform the pullover maneuver based on the list of qualities for the region; and maneuvering, by the

one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*);

receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device, determining a pickup area for the ride service request, along with a loading point that is within a pickup area, causing the motion planning system to determine a route from a starting location of the AV to the pickup area, causing the motion control system to move the AV along the route toward the pickup area, (**See Dyer paragraph 0042;** "Although the client computing devices **420**, **430**, and **440** may each comprise a full-sized personal computing device, they may alternatively comprise mobile computing devices capable of wirelessly exchanging data with a server over a network such as the Internet. By way of example only, client computing device **420** may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device **430** may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph 0076;** " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices **110** may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger.");

determining a route from a starting location of the AV to the pickup area; (**See Dyer paragraph 0027;** "Planning system **168** may be used by computing devices **110** in order to determine and follow a route generated by a routing system **166** to a location. For instance, the routing system **166** may use map information to determine a route from a current location of the vehicle to a drop off location. ").

navigating the AV along the route toward the pickup area; (**See Dyer paragraph 0036;** "The computing devices **110** may control the vehicle in an autonomous driving mode by controlling various components. For instance, by way of example, the computing devices **110** may navigate the vehicle to a destination location completely autonomously using data from the detailed map information and planning system **168**. ");

Dyer does not explicitly teach but Kaufman teaches; before the AV reaches the pickup area, determining whether the processor has received a departure confirmation from the passenger electronic device, wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point, (**See Kaufman paragraph 0032;** "The AV coordination component 138 receives the information in the ride request, and in response to receiving such information, the AV coordination component 138 determines a route toward the pickup location of the person. In another example, the server coordination component 114 can determine the route and transmit the route to the autonomous vehicle 106. Responsive to the route being determined, the autonomous vehicle 106 travels along the route toward the pickup location of the person. As the autonomous vehicle 106 travels along the route, the autonomous vehicle 106 can ascertain pickup

refinement information for the person. In one example, pickup refinement information can be based on strength of a wireless signal emanating from the mobile computing device 104 as the autonomous vehicle 106 moves relative to the mobile computing device 104 (e.g., as the signal strength becomes stronger, the autonomous vehicle 106 can ascertain that it is getting closer to the person). In another example, the autonomous vehicle 106 can ascertain the pickup refinement information based upon a passenger identifier being displayed on the display 130 of the mobile computing device 104. In yet another example, the autonomous vehicle 106 can ascertain the pickup refinement information based upon a passenger identifier in the form of physical object being held in the air by the person. Other examples are also contemplated.”; also **see Kaufman paragraph 0052;** “The server computing system processes the ride request and transmits ride information to an autonomous vehicle that is available to pick up the person at the pickup location identified in the ride request. The mobile computing device can confirmation that the ride request has been accepted by the server computing system."); and if the one or more processors have received the departure confirmation, causing the motion control system to move the AV into the pickup area and stop at the loading point, otherwise causing the motion control system to either (a) move the AV to an intermediate stopping location before reaching the pickup area or to (b) cause the AV to pass through the pickup area; (**see Kaufman paragraph 0033;** "As the autonomous vehicle 106 continues to navigate proximate the pickup location identified in the ride request, the sensor (not shown) of the autonomous vehicle 106 scans the environment. When the autonomous vehicle 106 detects the passenger identifier based upon a sensor signal output by the sensor of the autonomous vehicle 106, the autonomous vehicle 106 ascertains identifying information based

upon the passenger identifier (e.g., the autonomous vehicle 106 reads a QR code and acquires the identifying information). The autonomous vehicle 106 compares the identifying information to the identifying information received from the server computing system 102. If the two pieces of identifying information do not match, the autonomous vehicle 106 continues to search for the passenger identifier. If the two pieces of identifying information match, the autonomous vehicle 106 maneuvers towards the person presenting the passenger identifier and allows the person entry into the autonomous vehicle 106").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Koufman request confirmation and passenger identifier. No new functionality would arise from the combination and the

combination would improve usability of Dyer by adding the confirmation and passenger identifier. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 11, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer does not teach but Kaufman teaches, wherein the instructions to determine the loading point comprise instructions to: determine a passenger identifier that is associated with the passenger electronic device; access a passenger profile server and query the passenger profile server to return a passenger profile that corresponds to the passenger identifier; access

the returned passenger profile to extract one or more loading point location rules from the returned passenger profile; and use the loading point location rules to select the loading point from a set of candidate loading points (**see Kaufman paragraph 0026;** "The server computing system 102 is configured to receive a ride request from the mobile computing device 104, wherein the ride request comprises an identifier for a passenger account of a person and ride information, and further wherein the ride information can include a pickup time, a pickup location, a ride destination, a number of passengers who are to be included in the ride, etc. In response to receiving the ride request, the server computing system 102 is configured to transmit a signal to the autonomous vehicle 106, wherein the signal comprises a ride instruction. The ride instruction can indicate at least some of the content of the ride request, including the ride information and a passenger identifier that is assigned to the person. The passenger identifier may be a barcode (e.g., a QR code), a pattern or a sequence of patterns, a color or a sequence of colors, a key chain (with unique shape, color, a wireless transceiver, etc.). In this example, the autonomous vehicle 106 conducts a search for the passenger identifier while the autonomous vehicle 106 travels through a search area that is proximate to the pickup location identified in the ride request."; as well **as see Kaufman paragraph 0054;** "With reference now to FIG. 5, an exemplary methodology 500 performed by a server computing system is illustrated, wherein the server computing system is configured to instruct an autonomous vehicle to pickup a person and transport the person to a desired destination of the person. The methodology 500 starts at 502, and at 504 the computing system receives a ride request from a mobile computing device, wherein the ride request comprises an identifier of the person, a pickup location, and a destination. At 506, the server computing system

transmits ride instructions to an available autonomous vehicle from among a fleet of autonomous vehicles that are in network communication with the server computing system. The ride instructions can include, for instance, identifying information for the person (which is assigned to a passenger identifier), a pickup location, a pickup time, a destination location, etc. The server computing system can identify the autonomous vehicle based upon preferences in a passenger account of the person, based upon a number of riders indicated in the ride request, etc.").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Koufman confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 12, Modified Dyer teaches, the ride service system of claim 11 as discussed above. Dyer further teaches, wherein the loading point location rules comprise one or more rules to prioritize one or more of the following characteristics to determine the loading point: a lane segment that is in front of an entry of a building having a location that corresponds to a location of the passenger electronic device; a lane segment that is midway between two cross streets; or a lane segment that is as close as possible to the cross street that is next in the

direction of traffic while remaining a threshold distance away from that cross street; (**see Dyer paragraph 0073 – 0075**; "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.

In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).

In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities or as some form of additional cost in the ranking. For example. if a region (such as a block of a street)

typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 13, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer further teaches, further comprising additional programming instructions that are configured to cause the processor to determine the intermediate stopping location as a location that: is along the route between a current location of the AV and the loading point; and satisfies each of a plurality of permissible stopping location criteria; (**See Dyer paragraph 0075;** "In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities *(Examiner notes; list of qualities can also be a plurality of permissible stopping location criteria)* or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that

region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 14, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer further teaches, further comprising additional programming instructions that are configured to cause the processor to, upon receipt of the departure confirmation: before the AV reaches the loading point, process data received from a perception system of the AV to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV; and only stop the AV at the loading point to perform the loading operation if the loading operation will not obstruct the another vehicle, otherwise cause the AV to navigate to an intermediate stopping location and notify the passenger electronic device that the loading operation will be performed at the intermediate stopping location; (**see Dyer paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)."; also **see Dyer paragraph 0073;** " In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a

region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.").

Regarding claim 15, Modified Dyer teaches, the ride service system of claim 14 as discussed above. Dyer further teaches, wherein the instructions to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV comprise instructions to: determine whether the loading point is in a location that will require the AV to at least partially block a lane in the other vehicle that is following the AV is currently traveling; and determine that the AV will obstruct the other vehicle if: the loading point is in a location that will require the AV to at least partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop; (**see Dyer paragraph 0033;** "The perception system 172 also includes one or more components for detecting objects external to the vehicle such as other vehicles, obstacles in the roadway, traffic signals, signs, trees, etc.…."; **also see Dyer paragraph 0045;** "Other signals stored in the storage system 450 may also be determined by the server computing devices and/or human operators for areas around such pullover locations, or the regions discussed above. These other signals may include the number of available pullover

locations (as the vehicle drives through the region), how congested such areas are, how many other vehicles are pulling over in the region (as the vehicle drives through the region), how many vehicles are double parked within the region, whether there are other types of road users (such as pedestrians or cyclists) within the region, etc."; further **see Dyer paragraph 0046;** "Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle *(Examiner notes- tracking delta between the other road user's that would include vehicles behind the AV)*. This vehicle inconvenience value may be determined from prior attempts to pullover which may have caused problems to other road users *(Examiner notes – pullover causing problems to other users can include: partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop)* .For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience).

Regarding claim 16, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer doesn't teach but Kaufman teaches, further comprising additional programming instructions that are configured to cause the processor to, after receiving the departure confirmation and before the AV reaches the loading point, processing data received from the perception system of the AV to: (**see Kaufman paragraph 0055;** "At 508, the server computing system receives confirmation from the autonomous vehicle that the ride instructions have been received. At 510, the server computing system transmits a passenger identifier (e.g., a barcode,

a color, a pattern, etc.) to the mobile computing device if the mobile computing device does not already have the passenger identifier. In addition, the server computing system can transmit instructions to the mobile computing device, wherein the instructions indicate when the person is to present the passenger identifier and a direction that the passenger identifier is to be presented. At 512, the server computing system receives an indication that the autonomous vehicle has successfully located the passenger *(Examiner notes; location of the passenger is equivalent to that receiving the confirmation of departure)* and the ride has begun. The methodology 500 completes at 510." Also **see Kaufman paragraph 0039**; "the computing system **214** includes a perception system **220** that can perceive objects within an environment of the autonomous vehicle **106**.");

identify one or more objects located proximate to the loading point; measure a length of a stop zone that includes the loading point and that is free from obstruction; use a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while complying with permissible stopping criteria; and only stop at the loading point to perform the loading operation if the AV can non-suddenly stop within the stop zone while complying with the permissible stopping criteria, otherwise navigate the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location; (**see Kaufman paragraph 0010;** "In another example, the person may not have ordered a ride through the application (e.g., the person's mobile telephone is dead, or the person's mobile telephone does not get service). The person leaves the basketball arena and presents his or her mobile telephone to an autonomous vehicle that is not currently providing a ride, is not in route to pick

up a person who has hailed a ride, and/or which lacks capacity for the person (e.g., the autonomous vehicle already has several riders). The autonomous vehicle can detect the passenger identifier for the person from among the other visual stimuli in the surrounding environment based upon a sensor system output by a sensor of the autonomous vehicle. The computing system of the autonomous vehicle can identify a passenger account based upon the passenger identifier, and the autonomous vehicle may then stop to pick up the person and transport the person to a destination location indicated by the person."; also **see Kaufman paragraph 0033;** "As the autonomous vehicle 106 continues to navigate proximate the pickup location identified in the ride request, the sensor (not shown) of the autonomous vehicle 106 scans the environment. When the autonomous vehicle 106 detects the passenger identifier based upon a sensor signal output by the sensor of the autonomous vehicle 106, the autonomous vehicle 106 ascertains identifying information based upon the passenger identifier (e.g., the autonomous vehicle 106 reads a QR code and acquires the identifying information). The autonomous vehicle 106 compares the identifying information to the identifying information received from the server computing system 102. If the two pieces of identifying information do not match, the autonomous vehicle 106 continues to search for the passenger identifier. If the two pieces of identifying information match, the autonomous vehicle 106 maneuvers towards the person presenting the passenger identifier and allows the person entry into the autonomous vehicle 106").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective

filing date of present invention to modify Dyer identification of pullover region with Koufman request confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the request confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable;

Regarding claim 17, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer further teaches, further comprising additional programming instructions that are configured to cause the processor to, after the AV reaches the pickup area and in response to receiving a nudge request: process data received by the perception system of the AV to determine whether any objects would obstruct the AV if the AV would move to a position that is ahead of the loading point by a specified distance; and if no objects would obstruct the AV, update the loading point to move forward by the specified distance and move the vehicle to the updated loading point; (**see Dyer paragraph 0046;** "Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. Such signals may include how long passengers take to arrive, board and depart at a pullover location, the passenger inconvenience value of a pullover location, the vehicle inconvenience at pullover location, how long the vehicle is able to stay in a pullover location, etc. The values for passenger inconvenience and vehicle inconvenience may be determined, for instance, on a scale of 0 to 1 using a model which generates such values given map data and sensor data collected during

such attempted pullovers. Passenger inconvenience values may represent how convenient a particular pickup up or drop off was for a passenger by measuring how much extra distance is imposed on the passenger by the selection of a particular pullover location. Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle. This vehicle inconvenience value may be determined from prior attempts to pullover which may have caused problems to other road users. For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience value.").

Regarding claim 18, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer further teaches, further comprising additional programming instructions that are configured to cause the one or more processors to, in response to not receiving the departure confirmation, determine the intermediate stopping location by: selecting a road segment that is a nearest useable road segment to the pickup location; identifying, from one or more permissible stopping points along the road segment, the permissible stopping point that is closest to the pickup location; and using the permissible stopping point that is closest to the pickup location as the intermediate stopping location; (**See Dyer Paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a

passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)." ; **also see Dyer paragraph 0046;**"Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. "; as well **see Dyer paragraph 0048;** "In order to build a model, the server computing devices 410 may access the signals of the storage system 450 described above. Using these signals or values determined from these signals, a model that identifies expected characteristics of regions may be built by the server computing devices 410. The model may be trained such that for a given geographic constraint (such as a specific location or a region) and time constraint (such as day of the year, calendar month, day of week, and/or time of day), the model may provide a list of expected qualities for a region. In this regard, if a specific location is provided, the region may correspond to a region that includes the specific location. In this regard, the signals or values determined from these signals for a given region where the sensor data corresponding to the signals was collected may be used as training outputs for the model, and the specific location or given region and a time when the sensor data was collected may be used as training inputs to the model. In the case of a machine-learned model, the training may essentially tune parameter values for the model.").

Regarding claim 19, Dyer teaches, A method of determining a stopping location for a vehicle before picking up a passenger; (**See Dyer paragraph 002;** "determine, by the one or more

processors, whether to attempt to find a pullover location within the region to perform the pullover maneuver based on the list of qualities for the region; and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*);

the method comprising: by one or more processors, at least one of which is a component of an autonomous vehicle (AV): (**See Dyer paragraph 002;** "One aspect of the disclosure provides a method of maneuvering a vehicle in an autonomous driving mode. The method includes identifying, by the one or more processors, a time constraint for a pullover maneuver; identifying, by the one or more processors, a geographic constraint for the pullover maneuver; inputting, by the one or more processors, the time constraint and the geographic constraint into a model in order to receive a list of qualities for a region, the region including a plurality of pullover locations; determine, by the one or more processors, whether to attempt to find a pullover location within the region to perform the pullover maneuver based on the list of qualities for the region; and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*);

receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device; determining a pickup area for the ride service request, along with a loading point that is within a pickup area; determining a route from a starting location of the AV to the pickup area; (**See Dyer paragraph 0042;** "Although the client computing devices **420**, **430**, and **440** may each comprise a full-sized personal computing device, they may

alternatively comprise mobile computing devices capable of wirelessly exchanging data with a server over a network such as the Internet. By way of example only, client computing device **420** may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device **430** may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph 0076;** " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices **110** may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger.");

determining a route from a starting location of the AV to the pickup area; (**See Dyer paragraph 0027;** "Planning system **168** may be used by computing devices **110** in order to determine and follow a route generated by a routing system **166** to a location. For instance, the routing system **166** may use map information to determine a route from a current location of the vehicle to a drop off location. ").

navigating the AV along the route toward the pickup area; (**See Dyer paragraph 0036;** "The computing devices **110** may control the vehicle in an autonomous driving mode by controlling various components. For instance, by way of example, the computing devices **110** may navigate the vehicle to a destination location completely autonomously using data from the detailed map information and planning system **168**. ");

determining an intermediate stopping location as a location that (a) is along the route between a current location of the AV and the loading point, and (b) satisfies each of a plurality of permissible stopping location criteria; (**See Dyer paragraph 0075;** "In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities *(Examiner notes; list of qualities can also be a plurality of permissible stopping location criteria)* or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

navigating the AV along the route toward the pickup area; before the AV reaches the pickup area, (**See Dyer paragraph 0036;** "The computing devices **110** may control the vehicle in an autonomous driving mode by controlling various components. For instance, by way of example, the computing devices **110** may navigate the vehicle to a destination location completely autonomously using data from the detailed map information and planning system **168**. ");

Dyer does not explicitly teach but Kaufman teaches, determining whether the one or more processors have received a departure confirmation from the passenger electronic device, wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point; (**See Kaufman paragraph 0032;** "The AV coordination component 138 receives the information in the ride request, and in response to receiving such information, the AV coordination component 138 determines a route toward the pickup location of the person. In another example, the server coordination component 114 can determine the route and transmit the route to the autonomous vehicle 106. Responsive to the route being determined, the autonomous vehicle 106 travels along the route toward the pickup location of the person. As the autonomous vehicle 106 travels along the route, the autonomous vehicle 106 can ascertain pickup refinement information for the person. In one example, pickup refinement information can be based on strength of a wireless signal emanating from the mobile computing device 104 as the autonomous vehicle 106 moves relative to the mobile computing device 104 (e.g., as the signal strength becomes stronger, the autonomous vehicle 106 can ascertain that it is getting closer to the person). In another example, the autonomous vehicle 106 can ascertain the pickup refinement information based upon a passenger identifier being displayed on the display 130 of the mobile computing device 104. In yet another example, the autonomous vehicle 106 can ascertain the pickup refinement information based upon a passenger identifier in the form of physical object being held in the air by the person. Other examples are also contemplated."; also **see Kaufman paragraph no 0052;** "The server computing system processes the ride request and transmits ride information to an autonomous vehicle that is available to pick up the person at the pickup location

identified in the ride request. The mobile computing device can confirmation that the ride request has been accepted by the server computing system.");

and in response to determining that the system has received the departure confirmation: causing the AV to navigate toward the pickup area, identifying one or more objects located proximate to the loading point, measuring a length of a stop zone that includes the loading point and that is free from obstruction, using a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, and stopping at the loading point to perform the loading operation if the system determines that the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location; (**see Kaufman paragraph 0010;** "In another example, the person may not have ordered a ride through the application (e.g., the person's mobile telephone is dead, or the person's mobile telephone does not get service). The person leaves the basketball arena and presents his or her mobile telephone to an autonomous vehicle that is not currently providing a ride, is not in route to pick up a person who has hailed a ride, and/or which lacks capacity for the person (e.g., the autonomous vehicle already has several riders). The autonomous vehicle can detect the passenger identifier for the person from among the other visual stimuli in the surrounding environment based upon a sensor system output by a sensor of the autonomous vehicle. The computing system of the autonomous vehicle can identify a passenger account based upon the passenger identifier, and the autonomous vehicle

may then stop to pick up the person and transport the person to a destination location indicated by the person."; also see Kaufman see **Kaufman paragraph 0033;** "As the autonomous vehicle 106 continues to navigate proximate the pickup location identified in the ride request, the sensor (not shown) of the autonomous vehicle 106 scans the environment. When the autonomous vehicle 106 detects the passenger identifier based upon a sensor signal output by the sensor of the autonomous vehicle 106, the autonomous vehicle 106 ascertains identifying information based upon the passenger identifier (e.g., the autonomous vehicle 106 reads a QR code and acquires the identifying information). The autonomous vehicle 106 compares the identifying information to the identifying information received from the server computing system 102. If the two pieces of identifying information do not match, the autonomous vehicle 106 continues to search for the passenger identifier. If the two pieces of identifying information match, the autonomous vehicle 106 maneuvers towards the person presenting the passenger identifier and allows the person entry into the autonomous vehicle 106").

Both Dyer and Kaufman are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pul lover region with Koufman request confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the request confirmation, passenger identifier and ride instructions. Further, finding that one of

ordinary skill in the art would have recognized that the results of the combination were predictable;

Regarding claim 20, Modified Dyer teaches, the method of claim 19 as discussed above. Dyer further teaches, further comprising, by the one or more processors after determining that the system has received the departure confirmation: before the AV reaches the loading point, processing data received from a perception system of the AV to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV; and only stopping the AV at the loading point to perform the loading operation if the loading operation will not obstruct the another vehicle, otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location; (**see Dyer paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)."; also **see Dyer paragraph 0073;** " In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked

region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.").

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LIDIA KWIATKOWSKA whose telephone number is **(571)272-5161**. The examiner can normally be reached Monday-Friday 8:00-5:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **Anne Antonucci** can be reached on **(313)446-6519**. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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